CS4641/CS7641 Homework 3 — Non-Programming

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Deadline: November 20th, 11:59 pm ET

- No unapproved extension of the deadline is allowed. Refer to the syllabus for the late submission policy.
- Any section marked "Bonus for Undergrads" is optional extra credit for undergrads, but is **required for graduate students!** Any section marked "Bonus for All" is optional extra credit for everyone.
- Poorly formatted work may disrupt grading and may result in a **0 on affected questions**. See "Completing the Assignment" in the instructions below for more details.
- On most questions, work is required. Failure to show your work when expected may result in a **0 on affected questions**. See "Showing your Work" in the instructions below for more details.
- Failure to properly assign pages in Gradescope may disrupt grading and may result in a **0 on affected questions**. See "Submitting your Work" in the instructions below for more details.
- Discussion is encouraged! You are welcome to discuss at a whiteboard level with peers, but each student must write their own answers and explicitly mention any high-level collaborators. Plagiarism is a **serious offense**. You are not allowed to copy and paste, or paraphrase, or submit materials created or published by others, as if you created the materials. A good heuristic for cooperation is to discuss the questions together without discussing specifics, then write up your work and conclusions separately. If a conversation with a peer was meaningfully useful, please add their name to the Collaborators list below.
- Working with a generative-AI platform **may constitute plagiarism**. In line with the Joyner Heuristic being used by many classes at GT, you should treat collaboration with generative-AI as collaboration with a knowledgeable peer. It would be appropriate to ask about topics and formulae, but inappropriate to share the question and ask them to solve it. **Sharing the question or your work verbatim with an AI agent so as to generate an answer to the question is considered academic dishonesty and will be treated the same as any other incident of academic dishonesty.** If you're stuck, make a post on Ed, or visit office hours!
- Even using generative-AI for formatting your answers is **not permitted**. While we understand that LaTeX has a learning curve, you may not use any generative-AI platform to improve your writing or LaTeX formatting. These tools inherently produce output that may include a partial or complete solution to the question, including corrections to work you give it, even if purely prompted for syntactic use. Additionally, you have no custody over the data you give these platforms, which may leak or be used to inform subsequent training. Thus, while you are more than welcome to ask it about LaTeX commands and formatting tips, **sharing the question or your answer to a question verbatim with an AI agent, even purely for syntactic use, is considered academic dishonesty and will be treated the same as any other incident of academic dishonesty.** If you need formatting help, make a post on Ed, or visit office hours!
- All incidents of suspected dishonesty, plagiarism, or violations of the Georgia Tech Honor Code will be subject to the institute's Academic Integrity procedures. If we observe any (even small) issues detected by Gradescope or our TAs, **WE WILL DIRECTLY REPORT ALL CASES TO OSI**, which may, unfortunately, lead to a very harsh outcome, for example, academic probation or dismissal, a 0 grade for assignments concerned, and prohibition from withdrawing from the class.

Collaborators: *Please include whiteboard-level collaborators here, if any.*

Instructions

Completing the Assignment

- We provide you the source .tex files that contain the questions. We recommend completing the non-programming part of the assignment in a LaTeX editor. The ability to use LaTeX is a valuable skill in academia and in industry, and it's generally a clean way to render complicated math. Make sure that your work is formatted correctly. For example, you must submit $\sum_{i=0} x_i$ instead of sum_{i=0} x_i. A useful guide on LaTeX is provided in this Edstem post.
- You may use any LaTeX rendering tool you like! Overleaf is a great tool, with very few limitations. However, if you want finer control over packages and faster rendering, you may want to download a TeX distribution, a perl binary (if you're on Windows), and the "Latex Workshop" extension in VSCode to get live rendering in VSCode.
- If you're uncomfortable with using LaTeX, no worries! You can simply use the pdf version of the non-programming assignment and build your answers in any word or markdown processor you prefer. However, there are some caveats:
  1. **We will NOT grade handwritten work.** You must use some digital processor to render your work into a cleanly formatted pdf.
  2. It is **your responsibility** to ensure the questions are correctly numbered. While we will attempt to grade your work accurately, misnumbering may cause misinterpretation.
  3. It is **your responsibility** to ensure you're answering the question as it is written. We find that students who work without the questions embedded in their document tend to misinterpret the question more often. If you work in a separate document, read carefully and consider copying over the source question, even if its formatting may be lost.

Showing your Work

- When a question asks you to show your work, you must provide a justification for your solution. This may include calculations or text-based explanation. Though basic arithmetic may be combined, your work should be thorough and clear enough for a reader to follow your line of reasoning.
- When a question asks you to prove or provide a proof of some statement, you must provide a formal proof. These questions have a higher burden of work, and must constitute a valid inference. Again, while basic arithmetic can be combined, your work should be defensible as a formal proof. Any gaps in logic or unstated assumptions may result in deductions. Be careful using identities you may find online, since they may assume the fact you are trying to prove.
- If not otherwise indicated, no justification is needed, just the final answer.
- You will never be penalized for showing too much work, but you may be penalized for not showing enough.
- If you use an online resource, please cite it. We generally permit most resources, and so, citing it will only be to your benefit in the case of a plagiarism investigation (in the common case where multiple students utilize the same source).

Submitting your Work

- We will be using Gradescope for submission and grading of assignments. Complete the assignment, convert your assignment to a pdf, and submit it to the appropriate assignment denoted "Non-Programming" on Gradescope.
- When submitting your assignment on Gradescope, you are required to map pages of your pdf to subquestions on the homework. You should select a subquestion, then map it to **every** page which contains any portion of your solution to that subquestion.
- **You must map every page on which you have content that you would like be graded.** During the grading of each particular subquestion, we only see the pages you assign to that subquestion, so you must assign everything you would like us to grade.
- **Improperly mapped questions will receive penalties, up to and including 0 credit for the question.** You are permitted to submit a regrade request in this event; however, the review of such a request is at the sole discretion of the instructional staff and is not likely to be accepted.

- Some tips to correctly assign pages:
    1. Underassigning pages is a common mistake. For example, if your answer to question 4.2 starts halfway down on page 16, goes through page 17, and ends with a single line bleeding into page 18, assign pages 16, 17, *and* 18 to question 4.2. Assigning just page 16 may incur a page assignment penalty, or the grader may just interpret your work as incomplete.
    2. Keep in mind that overassigning is just as slowing to the grading process as underassigning is. Please try to keep your assignments tight around the content for the specific subquestion. E.g., assigning every page is just as useless as assigning no pages.

    Again, assign every page that has content relevant to your answer that you would like for us to see while grading. If you don't assign it, we can't see it.

Point Distribution

Q1: Profile Screening Using Naive Bayes [7pts]

Q2: Practical Application of PCA [5pts]

Q3: Ridge Regression [8pts]

Q4: Regression Under Noise [4.4% Bonus for All]

Points Totals:

- Non-Programming Base:
  - Undergrad: 20 pts
  - Graduate: 20 pts
- Non-Programming Bonus for Undergrad:
  - Undergrad: 0%
- Non-Programming Bonus for All:
  - Undergrad: 4.4%
  - Graduate: 4.4%
- For programming related points, see the Jupyter notebook.

1 Profile Screening Using Naive Bayes [7pts]

Naive Bayes Classifiers are discussed in section 5 of the Programming assignment, though to complete this section, you only need to know one formula. Based on some observed feature vector $x = [x_1, \ldots, x_d]$, Naive Bayes Classifiers choose a class, y, based on a simple argmax:

$$\arg\max_y \quad P(x_1 \mid y) \cdot \ldots \cdot P(x_d \mid y) \cdot P(y)$$

A tech company has devised an automated resume screening system (it's just Naive Bayes) for two distinct roles: Machine Learning Engineer and Product Manager. This system is designed to evaluate applicants based on five key binary attributes extracted from their resumes: {coding proficiency (1 for high, 0 for low), data analysis skills (1 for strong, 0 for weak), leadership experience (1 for yes, 0 for no), product design experience (1 for yes, 0 for no), marketing skills (1 for strong, 0 for weak)}.
They have compiled a dataset of 15 anonymized resumes from their past hire cycles, and would like to filter resumes from a career fair directly to the appropriate hiring teams.

Role	Coding	Data Analysis	Leadership	Product Design	Marketing
Machine Learning Engineer	1	1	0	0	1
Machine Learning Engineer	1	1	0	0	0
Machine Learning Engineer	1	0	1	0	0
Machine Learning Engineer	1	1	0	0	0
Machine Learning Engineer	0	1	0	0	0
Machine Learning Engineer	0	1	0	1	0
Machine Learning Engineer	0	0	0	0	0
Machine Learning Engineer	0	0	1	0	0
Product Manager	1	0	1	0	1
Product Manager	0	1	0	1	1
Product Manager	1	1	0	0	1
Product Manager	1	1	1	1	1
Product Manager	0	0	0	1	1
Product Manager	0	0	1	0	1
Product Manager	0	0	1	1	0

A new applicant's resume has been screened and binarized. They are proficient in coding and data analysis, they have no leadership nor product design experience, but they do have marketing skills. Using a Naive Bayes classifier, determine the most suitable job position for the new applicant. Show your work. Round the final probabilities to 6 decimal places. [7pts]

Warning

You must show show your intermediate values for the likelihoods, $P(x|y)$; the priors, $P(y)$; and the estimated logits, $\propto P(y|x)$; as queried from the given table. This is required work and will be graded, not just the final decision.

Also, don't divide your answer by the marginal, $P(x)$. Since this is not a function of the label, it will be the same for every label, thus this additional division is not necessary when using Naive Bayes classification and should be skipped. Doing so will result in a deduction during grading. You should use the formula presented.

2 Practical Application of PCA [5pts]

This question will ask about the application of PCA onto the dataset used in section 2 of the Programming assignment. While that section may be helpful, this question is standalone.

a. After applying PCA to the wine dataset, we can analyze what proportion of total variance is expressed by each component. By the nature of the algorithm, they actually already come sorted by how much variance they express. Usually, we keep the top N most expressive components, and we can add up the variance they capture for a cumulative graph. We can plot this for our 13-dimensional wine dataset.

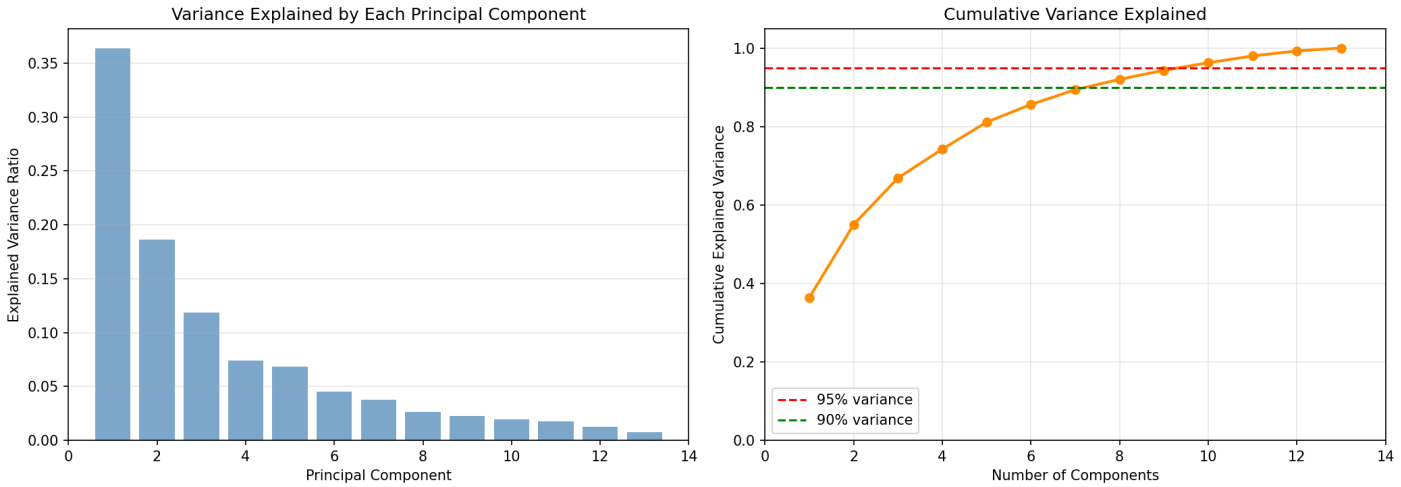


Figure 1: Left: the variance of each component, Right: the cumulative variance of the top N components together

Based on the cumulative variance plot above, what is the minimum number of principal components required to capture 95% of the total variance? What does this tell you about the redundancy in the original 13 features? [2pts]

b. PCA can be thought of as a projection onto a set of basis vectors that follow the directions of maximum variance in the data. PCA, then, is often a useful preprocessing step for classification models that perform simple inferences or computations on the features themselves, e.g., a decision tree, which repeatedly splits the dataset along individual features.

While it *often* does improve classification performance, PCA isn't *guaranteed* to do so. Why might choosing axes that maximize variance not necessarily improve class separability along the axes? [3pts]

3 Ridge Regression [8pts]

This question covers content from section 4.1-3 of the Programming assignment. While that section may be helpful, this question is standalone.

One weakness of linear regression is that it requires the pseudo-inverse of X, namely, $\theta = (X^\top X)^{-1} X^\top y$, which is problematic since highly correlated features may cause the inverse of $X^\top X$ to not exist or be numerically instable. Ridge regression avoids this by adding a regularization term to the solution: $\theta = (X^\top X + \lambda I)^{-1} X^\top y$, for some $\lambda > 0$.

Prove that, for any X, $X^\top X + \lambda I$ is invertible (with $\lambda > 0$). [8pts]

Hint

There are several ways to attack this proof. However, if you're unsure where to begin, we'll give a starting point. A useful thing to note is that every "positive definite" matrix is invertible, and it's much easier to prove that $X^\top X + \lambda I$ is "positive definite." We say a matrix A is positive definite if and only if $v^\top A v > 0$ for all vectors $v \neq \mathbf{0}$.

4 Regression Under Noise [4.4% Bonus For All]

Consider a linear model of the form:

$$\hat{y}(\vec{x_i}, \theta) = \theta_0 + \sum_{d=1}^{D} \theta_d x_{id}$$

where $\vec{x_i} = (x_{i1}, \ldots, x_{iD}) \in \mathbb{R}^D$ and weights $\theta = \{\theta_0, \ldots, \theta_D\} \in \mathbb{R}^{D+1}$. Given the the D-dimensional input data set $X = \{\vec{x_1}, \ldots, \vec{x_N}\}$ with corresponding target values $y = \{y_1, \ldots, y_N\}$, the sum-of-squares error function for the model is defined as:

$$E_D(\theta) = \frac{1}{2} \sum_{n=1}^{N} [\hat{y}(\vec{x_n}, \theta) - y_n]^2$$

Now, suppose that Gaussian noise $\vec{\epsilon_n} \in \mathbb{R}^D$ is added independently to each of the input samples $\vec{x_n}$ to generate a new data set $X' = \{\vec{x_1}', \ldots, \vec{x_N}'\} = \{\vec{x_1} + \vec{\epsilon_1}, \ldots, \vec{x_N} + \vec{\epsilon_N}\}$. To be specific, $\vec{x_k}' = (x_{k1} + \epsilon_{k1}, \ldots, x_{kD} + \epsilon_{kD})$. We define the Gaussian noise to be zero mean (i.e., $E[\epsilon_{ij}] = 0$), with variance σ^2 (i.e., $\mathsf{Var}(\epsilon_{ij}) = \sigma^2$), and each individual noise term is independent and identically distributed (i.i.d.).

a. Show that $\hat{y}(\vec{x_n}', \theta) = \hat{y}(\vec{x_n}, \theta) + \sum_{d=1}^{D} \theta_d \epsilon_{nd}$. Show your work. [1.0% Bonus For All]

b. Denote the sum-of-squares error function of the noised data set X' as $E_D(\theta)'$. Show that the expectation of $E_D(\theta)'$ is equivalent to the sum-of-squares error for noise-free input data, $E_D(\theta)$, with the addition of some term which resembles the ℓ_2 norm of θ, in which the bias parameter θ_0 is omitted, that is, a "weight-decay regularizer." In other terms, you need to show that

$$\mathbb{E}[E_D(\theta)'] = E_D(\theta) + \text{Regularizer}$$

Show your work. [3.4% Bonus For All]

Hint

Since the noise has $\mathsf{Var}(\epsilon_{ij}) = \sigma^2$ and zero mean, we can see:

$$\begin{aligned} \mathsf{Var}(\epsilon_{ij}) &= \mathbb{E}[\epsilon_{ij}^2] - (\mathbb{E}[\epsilon_{ij}])^2 \\ \sigma^2 &= \mathbb{E}[\epsilon_{ij}^2] - 0^2 \\ \mathbb{E}[\epsilon_{ij}\epsilon_{ij}] &= \sigma^2 \end{aligned}$$

Also, since the noise terms are i.i.d., we can say something about the product of different noise terms.

$$\begin{aligned} &\mathbb{E}[\epsilon_{ij}\epsilon_{mn}] = \mathbb{E}[\epsilon_{ij}]\mathbb{E}[\epsilon_{mn}] \qquad \text{by the definition of independence, iff } i \neq m \vee j \neq n \\ &\mathbb{E}[\epsilon_{ij}\epsilon_{mn}] = 0 \cdot 0 = 0 \end{aligned}$$

Altogether, the expectation of the product of two noise terms is piecewise, depending on whether they are the same noise term.

$$\mathbb{E}[\epsilon_\phi \epsilon_\psi] = \begin{cases} \sigma^2 & \phi = \psi \\ 0 & \phi \neq \psi \end{cases}$$